TECHNICAL SPECIFICATION FOR PORTABLE THREE PHASE FULLY AUTOMATIC TEST SYSTEM WITH INTEGRATED CURRENT AND VOLTAGE SOURCE OF ACCURACY 0.05

1.0 SCOPE:

This specification covers design, manufacture, testing, supply and delivery of PORTABLE THREE PHASE FULLY AUTOMATIC TEST SYSTEM WITH INTEGRATED CURRENT AND VOLTAGE SOURCE with accuracy class 0.05 in direct mode up to 120 Amp & class 0.2 with clamp on CT up to 120 Amp and necessary cables with other accessories for proper function of testing of single or Three phase Energy Meter of electro mechanical and electronic type of accuracy class 0.2, 0.5, 1.0 installed at the consumer's installations as well as in laboratory.

2.0 APPLICATION:

The function of the Portable Three Phase Fully Automatic Test System shall be suitable to generate the Voltage up to 500 Volts, Current up to 120 Amp and measure the system parameters. It shall be verify the accuracy of three phase or single phase energy meters in the laboratory as well as at site.

The Portable Three Phase Fully Automatic Test System shall be extensively used in field & laboratory for verification of the accuracy of all types of three phase whole current, LT-CT operated and single phase energy meter with help of Inbuilt Power Source.

3.0 SUPPLY VOLTAGE:

The Portable Three Phase Fully Automatic Test System shall have facility to power up from auxiliary power supply from range of 85...265V, 47…65 Hz.

4.0 SERVICE (CLIMATIC) CONDITIONS:

The equipment to be supplied against this specification should be capable of performing and maintaining the required accuracy for satisfactory operation under all tropical conditions mentioned below:

a)	Maximum ambient temperature (degree C)	:	50
b)	Maximum ambient temperature in shade(degree C)	:	45
c)	Minimum temperature of air in shade(degree C)	:	3.5
d)	Maximum daily average temperature(degree C)	:	40
e)	Maximum yearly weighted average temperature(degree C):32		
f)	Relative Humidity (in %)	:	10 to 95
g)	Maximum Annual rainfall(mm)	:	1450
h)	Maximum wind pressure (Kg/m. sq)	:	150
i)	Maximum altitude above mean sea level (meters)	:	1000
j)	Isoceraunic level (days/year)	:	50
k)	Seismic level (Horizontal acceleration)	:	0.3g

5.0 STANDARDS APPLICABLE:

Unless otherwise Specified elsewhere in this Specification, the Portable Three Phase Fully Automatic Test System shall conform to relevant clause of the following standards in all respects including performance and testing thereof to the following Indian /international Standards to be read with up to-date and latest amendments / revision thereof:

IS 15707	:	Testing, evolution, installation and maintenance of AC electricity meters- code of practice
IS 12346	:	Testing equipment for AC Electrical Energy Meters
IEC 60736	:	Testing equipment for electrical energy meters.

6.0 GENERAL REQUIREMENTS:

a. All the materials, electronic and power components and ICs used in the manufacturing of the equipment shall be of highest quality and reputed make to ensure higher reliability, longer life and sustained accuracy.
b. The electronic components shall be mounted on the PCB using latest surface mount technology (SMT).
c. The Portable Three Phase Fully Automatic Test System shall be of rugged construction, lightweight and shall be portable and handy. It shall have ergonomic design.
d. The equipment shall have facility to interchange the clamp on CT from one equipment to equipment without affecting the overall accuracy.
e. A set of voltage & current leads with suitable connectors shall be supplied along with the Portable Three Phase Fully Automatic Test System.
f. An error calculator shall be incorporated in the Portable Three Phase Fully Automatic Test System, which shall have facility to calculate error in percentage of meter under test by feeding the meter constant and number of pulses for which meter was tested with Electronic Reference Standard Meter in Portable Three Phase Fully Automatic Test System, through the Inbuilt key board.
g. The equipment shall have inbuilt legible LCD/TFT display with front panel keys for its operation and better integrity of the equipment. Operation only through external device or pc/laptop is not acceptable.
h. The Equipment shall be compatible for carrying out Accuracy test, Voltage & Frequency variation test, Harmonics injection and Voltage Unbalance tests as per IEC/IS standards.
i. The Equipment shall have facility to test energy meter in Fully Automatic condition without using of any external PC/Laptop.

7.0 GENERATION RANGES:

7.1 Voltage

Range	:	20V to 500V (P-N)
Setting accuracy	:	Better than 0.2%

Distortion	:	< 0.5%
Power	:	30 VA
Harmonics	:	up to 40th
Harmonic Amplitude	:	max. 40 % @ 2nd - 10th .
		max. 10 % @ 11th. – 40th.

7.2 Current

Range	:	1 mA to 120 A for each phase
Setting accuracy	:	Better than 0.2%
Distortion	:	< 0.5%
Power	:	60 VA
Harmonics	:	up to 40th
Harmonic Amplitude	:	Max. 40 % @ 2nd. – 10th.
		Max. 10 % @ 11th. – 40th.

7.3 Phase Angle

Range	:	0° to 359.99°
Accuracy	:	better than 0.1° (at symmetrical load)
Resolution	:	0.01°

7.4 Frequency

Range	:	45 to 65 Hz (quartz controlled) 50 / 60 mains synchronized
Accuracy	:	better than 0.01 Hz
Resolution	:	0.01 Hz

8.0 MEASUREMENT RANGES:

The Portable Three Phase Fully Automatic Test System shall have following measurement ranges to carry out the testing of single phase and three phase energy meters.

Test Voltage Range	5 mV to 500V (Phase to Neutral)
Test Voltage resolution	0.01 V

Test Voltage Accuracy	0.03 % (30V……500V)
Test Current range	1mA to 120A in direct and Clamp (It should start sensing the current from on CT mode.1 mA in Direct Mode and 5mA in Clamp-on CT mode)
Test Current resolution	0.01mA to 0.0001A
Test Current accuracy	Direct Mode < 0.025 % (10 mA ... 120 A) <0.2 % (5 mA … <10 mA) Clamp on CT Mode < 0.15 % (500 mA ... 120 A)
Phase Angle measurement range	0 to 360°
(It shall measure PF with the resolution of 0.0001)	
Phase Angle Error	0.01°
Frequency range	45 to 65 Hz
Resolution	0.01 Hz
Power/energy measurement error	0.05% (10 mA to 120A) Direct mode
(For active and reactive measurement)	0.2% (500mA to 100A) Clamp OnCT
Power/energy measurement temperatures drift	< 25 PPM/K

9.0 MEASUREMENT-MODE

The offered equipment should have following measurement modes to test meters of following type.

- Single Phase two wire active and reactive
- Three phase three wire active and reactive
- Three phase four wire active, reactive & apparent

10.0 ACCURACY REQUIREMENT:

The class of accuracy of the offered equipment in active and reactive power & energy measurement shall be 0.05% for the current range of 10 mA to 120A in direct mode measurement and 0.2% for the current range of 500 mA to 100A with Clamp on CT.

11.0 DISPLAY:

The offered calibrator shall have legible display of minimum “8” Color graphical back lit LCD display. External display is not acceptable.

The Portable Three Phase Fully Automatic Test System shall display the following system parameters namely:

- Phase to Phase voltage and Phase to neutral voltage for each phase
- Equipment shall have facility to measure and display active, reactive and apparent power (along with sign) of each phase. Equipment should have facility to display harmonics up to 40th order in tabular form, linear form, bar graph and Logarithmic form for voltage and current circuit with angle of superimposition.
- Harmonics distortion in each phase and total harmonic distortion.
- Phase current for each phase
- Phase angle between voltage
- Phase angle between voltage and current
- Continuous updating of energy (as per selected measuring mode) during error testing using scanner.
- Active, Reactive and apparent power of each phase & Total.
- Power factor of each phase & Total
- Frequency & Phase Sequence
- Error in Percentage

Offered equipment shall have facility to view/monitor system parameters, vector diagram and harmonics during performing the Error testing. This facility shall be provided without disturbing the running test so that any variation in system parameters can be observed by operator.

12.0 DISPLAY RESOLUTIONS:

Minimum resolutions-of-various parameters shall be as follows:

• Voltage	:	0.01 V
• Current	:	0.01mA to 0.0001A
• Power factor	:	0.001
• Energy		0.0001 (Wh)
• % Error Resolution	:	0.01

13.0 INSTALLATION CHECKING:

The Equipment shall be capable of indicating the following conditions by vector diagram and/or instantaneous values:

• Missing potential & Missing current.

• Reverse current if any current is reverse & Phase sequence "Forward or Reverse"

14.0 MEMORY:

The Portable Three Phase Fully Automatic Test System shall have facility to store up to 1000 error test results in its detachable memory card/stick along with following instantaneous parameters. The memory card/stick shall be situated at front panel of equipment.

- Voltage & Current of each phase
- Angle between voltage and current
- Power
- Measuring mode
- No of revolution
- MUT constant
- Error in percentage
- Energy logged/recorded by ERSS during test as per selected measurement mode while test was performed with scanner or snap switch.
- Test duration in hour, minute and seconds (with time of commencement of test and completion).
- Phase to phase voltage
- Distortion in voltage circuit.
- Distortion in current circuit.
- Phase angle between voltages
- Phase angle between current
- Selected current and voltage ranges
- Voltage ratio (if any) (As per meter under test rating plate)
- Current ratio (if any) (As per meter under test rating plate)
- Frequency in Hz
- Phase sequence
- Total power active, Reactive and apparent

The ERSS should also have facility to store:

- The error data up to at least 1000 tests, shall be stored in memory card/stick and these can be down loaded to computer by connecting it to PC so that print outs of test results can be taken out with compatible software.
- Harmonics in the form of table with absolute value of harmonics up to 40th order and angle of superimposition of harmonics.
- Harmonics in the form of bar graph.
- Equipment should have facility to store active, reactive and apparent powers due to harmonics along with sign (to show the direction of flow of harmonics).
- Wave forms : Two wave forms shall be stored together for analysis purpose
- Vector diagram with selected voltage and current ranges and measuring mode

15.0 INTERFACE:

The Portable Three Phase Fully Automatic Test System should have following interface provisions:

- Error compensated Clamp on CTs of class 0.2 (up to 120A).
- RS 232 Control with PC.
- Frequency output to calibrate the reference standard itself.

- The optical scanner head shall be capable to evaluate the error through calibrating pulses output of electronic meter & Red/Black mark on the rotor disc of electromechanical meter.
- Interface should be provided to control source through reference meter.
- VGA & Keyboard interface for service purpose.
- Memory card/Stick slot.

16.0 POWER CONSUMPTION:

The power consumption of the Three Phase Fully Automatic Test System at reference voltage, frequency, temperature and rated current shall not be more than 500 VA (rms).

17.0 CONSTRUCTIONAL FEATURES:

- The Power Source and Reference Standard Meter shall be integrated in single unit and shall accommodate in one ABS enclosure. The Enclosure should have facility of wheels so that Equipment can be ease carried out at field.
- The Equipment shall be provided with suitable leads of voltage and current connection. These leads shall be provided with various connectors to connect energy meter in different options / conditions.
- The Equipment shall be provided with Alfa- numeric keypad to select measuring mode, Enter MUT constant, number of revolution and to move the cursor on the screen. The operation of key pad is very user friendly just like key pad of mobile. External type keypad is not acceptable.
- The Equipment shall be designed in such a way that the weight of equipment is not more than 22 kg.
- The Equipment shall be provided with an optical sensor with suitable mounting arrangement so as to align the sensor properly with the LED impulse counter to sense, the LED impulses from the optical head of meter. The optical sensor along with the mounting arrangement shall be capable of being clamped to the meter body. The details drawing of mounting clamp to be submit by bidder along with bid.
- Indication for Power Supply and readiness of the equipment.
- Indication for status of each voltage and current supply.

18.0 FUNCTIONS

The equipment shall be used effectively for the following:

- Verification for accuracy of energy meters using scanner and using key pad start stop push button.
- It shall have facility to enter meter constant in imp/Kwh, imp/kVArh, imp/KVAh and imp/Wh, imp/VArh and Vah / imp.
- Offered equipment shall have facility to view/monitor system parameters, vector diagram and harmonics during performing the Error testing. This facility shall be

provided without disturbing the running test so that any variation in system parameters can be observed by operator

- It shall have facility to enter No. of pulses and revolutions of the meter under test. The no. of pulses and constant shall be entered up to 10 digits.
- It should have facility to display the energy logged during test while testing is performed by Scanner and snap switch.
- Harmonics analysis of the supply up to 40^{th} harmonics. Equipment should have facility to store active, reactive and apparent power along with sign (to show direction of flow) of each harmonics. Equipment should also have facility to show harmonics of each voltage and current with superimposition angle. Harmonics display should be in tabular form as well as in bar graph form.
- Waveform and vector display to analysis the circuit connections.
- Equipment shall have facility to operate in automatic mode without use of external PC.

19.0 ENCLOSURE:

The Power Source and Reference Standard Meter shall be integrated in single unit and shall accommodate in one ABS enclosure. Trolley Case shall be provided to ease of handling the Equipment in the field.

20.0 SOFTWARE:

The Portable Three Phase Fully Automatic Test System shall be supplied along with PC Software. The software shall be suitable for downloading the test results into IBM compatible PC using Card Reader and memory card/stick. The software shall have facility to generate the test report for individual testing and summary report of all test reports. Full Technical capability shall be demonstrated if asked by Purchaser. The software shall have facility to convert all stored test results in ASCII file format. The software shall be user friendly & menu driven.

21.0 ACCESSORIES:

Each Portable Three Phase Fully Automatic Test System shall be supplied along with following accessories:

i) Common optical sensor for automatic testing, which can be used to sense disc revolutions in electromechanical meters as well as indicating LED's in static meters.
ii) Mounting arrangement (clamp) for the optical sensor.
iii) Compensated Clamp on CT of class 0.2 up to 120A for on-line testing.
iv) 4 nos. of voltage lead with insulated clips. (1Red, 1Yellow, 1Blue, 1 Black)
v) a set of clips and connectors as following:
   - Cable Adopter / connection Pins 10Nos
   - Voltage adopters 4 Nos
   - Banana clips (straight) 6 Nos
   - Banana clips (bended) 4 Nos
   - Crocodile Clips 3 Nos.

- U clips 6 Nos.

vi) 6 nos. of Current leads to connect Electronic Reference Standard Meter in direct mode. (2 Red, 2 Yellow, 2 Blue)
vii) Serial communication cord with RS232 connector to control with PC.
viii) Base Computer Software (BCS)
ix) Operating Manual.

22.0 GUARANTEE:

The Portable Three Phase Fully Automatic Test System should be guaranteed for performance for a period of 60 (Sixty) Months from the date of commissioning or 66 (Sixty Six) Months from the date of receipt in stores, whichever date is earlier. The equipment found defective within the above guarantee period shall be repaired/ replaced by the supplier free of cost within Fifteen Days of receipt of intimation.

23.0 GUARANTEED TECHNICAL PARTICULARS:

A statement of guaranteed technical particulars shall be furnished in the format attached as per GTP along with the bid without which the bid shall be treated as non responsive.

24.0 AFTER SALES SUPPORT AND TRAINING:

The supplier shall arrange to provide free training for use of Portable Three Phase Fully Automatic Test System. The supplier shall provide competent and timely after sales service support.

25.0 PACKING AND FORWARDING:

The equipment shall be packed suitably for transport to withstand handling during transport. The supplier shall be responsible for any damage to the equipment during transit due to improper and inadequate packing. The easily damageable material shall be carefully packed and marked with appropriate caution symbol.

Each consignment shall be accompanied with a detailed packing list containing the following information.

i. Name of the consignee.

ii. Details of consignment.

iii. Destination.

iv. Handling and packing instructions.

v. Bill of material indicating contents of each package.

The packing shall be done as per the manufacturer's standard practice. However, the supplier should ensure the packing is such that, the material should not get damaged during transit by Rail/Road/Sea/Air.

26.0 DOCUMENTATION:

Each Portable Three Phase Fully Automatic Test System meter shall be supplied along with operating manual and Factory Test Report. One set of routine test certificate shall be supplied along with each electronic reference standard meter.

Guaranteed Technical Particulars for Portable Three Phase Power Source and Portable Reference Standard Meter (0.05)

S. No.	Item	Requirement	Remark.
1.	Manufacturers name and country	Please specify manufacturer name	
2.	Type of Meter	Please specify model number and attach product literature/brochure.	
3.	**POWER SOURCE**		
3.1	Output Voltage Range	20 -500V (P-N)	
3.2	Output Power in Voltage Circuit	30VA	
3.3	Harmonics Generation in Voltage	up to 40th	
3.4	Harmonic Amplitude	max. 40 % @ 2. - 10. max. 10 % @ 11. - 40.	
3.5	Angle range	0° to 359.9°	
3.6	Output Current Range	1 mA to 120 Amp	
3.7	Output Power in Current Circuit	60VA	
3.8	Harmonics Generation in Current	up to 40th	
3.9	Harmonic Amplitude	max. 40 % @ 2. - 10. max. 10 % @ 11. - 40.	
4.	**REFERENCE STANDARD METER**		
4.1	Accuracy class in power /energy measurement for active and reactive measurement.	Class 0.05 in direct mode and 0.2 with clamp on CT up to 120A for active, reactive and apparent energy measurement	
4.2	Voltage range	5mV… 500V AC	
4.3	Test Voltage Accuracy	<0.03 % (30V…..500V)	

S. No.	Item	Requirement	Remark.
4.4	Current range	1 mA to 120 A	
4.5	Test Current accuracy	Direct Measurement < 0.025 % (10 mA ... 120 A) <0.2 % (5 mA … <10 mA) Clamp on CT Measurement < 0.15 % (500 mA ... 120 A)	
4.6	Power/energy measurement error	0.05 in direct mode(10 mA to 120A) 0.2 in clamp on CT mode (from 500 mA to 100A)	
4.7	Power/energy measurement temperature drift	<25 PPM/K	
4.8	Frequency range	45 to 65 Hz	
5	Display	The offered calibrator shall have legible display of minimum 10.4" Color graphical back lit LCD display. External display is not acceptable. The Portable Three Phase Fully Automatic Test System shall display the following system parameters namely: - Phase to Phase voltage and Phase to neutral voltage for each phase - Equipment shall have facility to measure and display active, reactive and apparent power (along with sign) of each phase. Equipment should have facility to display harmonics up to 40th order in tabular form, linear form, bar graph and Logarithmic form for voltage and current circuit with angle of superimposition. - Harmonics distortion in each phase and total harmonic distortion. - Phase current for each phase - Phase angle between voltage - Phase angle between voltage and current	

S. No.	Item	Requirement	Remark.
		- Continuous updating of energy (as per selected measuring mode) during error testing using scanner. - Active, Reactive and apparent power of each phase & Total. - Power factor of each phase & Total - Frequency & Phase Sequence - Error in Percentage Offered equipment shall have facility to view/monitor system parameters, vector diagram and harmonics during performing the Error testing. This facility shall be provided without disturbing the running test so that any variation in system parameters can be observed by operator.	
6	Display resolution	• Voltage : 0.01 V • Current : 0.01mA to 0.0001A • Power factor : 0.001 • Energy : 0.0001 (Wh) • % Error Resolution : 0.01	
7	Memory	Minimum 1000 error test results.	
8	Features	- Equipment shall have facility to view/monitor system parameters during performing the Error testing without disturbing the running test. - Equipment shall have facility to interchange the clamp on CT from one equipment to other equipment without affecting the overall accuracy. - Equipment should have facility to test EUT automatically by using predefined test routines without external PC. - Waveform and vector display to analysis the circuit connections.	-
9	Functions	The equipment shall be used effectively for the following: • Verification for accuracy of energy meters using scanner and using key pad start stop push button. • It shall have facility to enter meter constant in imp/kwh, imp/kVArh, imp/kVah and Wh/imp, VArh/imp and Vah / imp. • Offered equipment shall have facility to	-

S. No.	Item	Requirement	Remark.
		view/monitor system parameters, vector diagram and harmonics during performing the Error testing. This facility shall be provided without disturbing the running test so that any variation in system parameters can be observed by operator • It shall have facility to enter No. of pulses and revolutions of the meter under test. The no. of pulses and constant shall be entered up to 10 digits. • It should have facility to display the energy logged during test while testing is performed by Scanner and snap switch. • Harmonics analysis of the supply up to 40th harmonics. Equipment should have facility to store active, reactive and apparent power along with sign (to show direction of flow) of each harmonics. Equipment should also have facility to show harmonics of each voltage and current with superimposition angle. Harmonics display should be in tabular form as well as in bar graph form. • Waveform and vector display to analysis the circuit connections. • Equipment shall have facility to operate in automatic mode without use of external PC.	
10	Carrying case:	Equipment shall be supplied with movable trolley for easy portability. The Power source and Reference Standard Meter shall be integrated in single unit and shall accommodate in one ABS enclosure	
11	Software	The Equipment shall be supplied along with PC Software which is suitable for downloading the test results into PC using RS 232 port. The software shall have facility to generate the test report for individual testing, summary report, sequential report for analysis and printing for test results for user defined parameters.	
12	Accessories	The Equipment shall be supplied along with	

S. No.	Item	Requirement	Remark.
		following accessories: i) Common optical sensor for automatic testing, which can be used to sense disc revolutions in electromechanical meters as well as indicating LED's in static meters. ii) Mounting arrangement (clamp) for the optical sensor. iii) Compensated Clamp on CT of class 0.2 up to 120A for on-line testing. iv) 4 nos. of voltage lead with insulated clips. (1Red, 1Yellow, 1Blue, 1 Black) v) a set of clips and connectors as following: • Cable Adopter / connection Pins10Nos • Voltage adopters4 Nos. • Banana clips (straight)6 Nos. • Banana clips (bended)4 Nos. • Crocodile Clips3 Nos. • U clips 6 Nos. vi) 6 nos. of Current leads to connect Electronic Reference Standard Meter in direct mode. (2 Red, 2 Yellow, 2 Blue) vii) Serial communication cord with RS232 connector to control with PC. viii) Base Computer Software (BCS) ix) Operating Manual.	